Death Avoidance as the Dominant Routing Signal: A Counterfactual Analysis on ScienceWorld

Anonymous ACL submission

Abstract

LLM agent routing (deciding when to rely on internal reasoning versus external tool use) is typically addressed by training routers directly, without first understanding what mechanistic signal drives the routing preference. This absence of diagnostic methods risks building routers atop confounded or trivial signal. We propose an episode-level counterfactual framework that compares pure-strategy outcomes (internal-only vs. deterministic-only) to decompose routing signal into its mechanistic components before any router is constructed. Applying this framework to ScienceWorld (11 task types, 245 episodes), we find that 78.9% of non-tie episode-level routing labels arise from death asymmetry—a finding replicated across three architecturally distinct 7–9B model families (spread 2.2 pp), revealing that in catastrophic-failure environments, signal diversity can collapse to a single dominant mechanism. Cross-environment probes on MATH and ALFWorld diagnose qualitatively different routing mechanisms (no extractable signal and information asymmetry, respectively), providing preliminary evidence of applicability beyond death-dominated settings. The framework distinguishes *methodological generality* (applicable to any environment with paired episode outcomes) from *empirical generality* (full quantitative validation on ScienceWorld only). A $\sim$50-episode diagnostic pilot provides preliminary build-vs-buy signal given large effect sizes ($d = 3.02$); environments with smaller effects may require larger samples.

1 Introduction

Large language models deployed as interactive agents must repeatedly decide *how* to compute: reason internally, or delegate to an external tool. In ScienceWorld (Wang et al., 2022), a text-based science simulation, internal reasoning is error-prone, as the model may hallucinate state changes (e.g., that water has boiled when the stove was never activated), while deterministic external computation eliminates such errors via exact state transitions. The practical question: *when should an agent route computation externally versus internally?*

Existing routing approaches frame this as a capability or efficiency decision: Toolformer (Schick et al., 2023) inserts API calls by perplexity reduction; ART (Paranjape et al., 2023) decomposes tasks into tool-augmented programs; Chameleon (Lu et al., 2023) selects heterogeneous modules per subtask; adaptive-computation methods route to different reasoning depths (Yang et al., 2026b,a), model tiers (Zhang et al., 2026c), or episode-level model selection under sparse rewards (Zhang et al., 2026a); concurrent analyses further presuppose diverse routing signals (Sun et al., 2026; Zhang et al., 2026b; Wu et al., 2026). These approaches, which operate in non-catastrophic settings, share the assumption that routing signal stems from diverse sources and a learned router can exploit it—an assumption that remains untested: no prior work has decomposed what drives routing decisions in interactive environments.

We propose a *diagnostic framework* comprising (i) episode-level counterfactual labeling under pure strategies (§2.2–2.3), and (ii) the Death Avoidance Factor (DAF), which decomposes the routing score gap into death-avoidance and non-death components (§2.4). Applied to ScienceWorld, **78.9% of non-tie routing labels arise from death asymmetry** (75 of 95), a penalty-independent finding. Under ScienceWorld's penalty (P=100), DAF reaches 90.4% (95% CI: $[81.1\%, 99.0\%]$). The internal strategy's hallucination-induced failures trigger catastrophic failure ("death," -100 penalty); once death episodes are removed, the gap collapses to $\Delta_{\text{nd}} = +2.84$ (§3.1).

This addresses a methodological gap: per-step oracle comparison degenerates under sparse rewards (97.4% tie rate; §2.2), and keyword-based heuristics fail structurally (§3.5). Our approach

runs complete episodes under pure-internal (ReAct (Yao et al., 2023b)) and pure-deterministic strategies, comparing episode outcomes.

Cross-context checks probe specificity: MATH Level 4–5 (N=795, code interpreter) shows no routing signal without catastrophic failure (73.5% vs. 73.0%, $p = 0.81$); Llama-3.1-8B replication (N=220) confirms death dominance across model families (direction consistency 10/10; §3.2). Four intervention axes (retry, Reflexion (Shinn et al., 2023), 10× scaling, action-space restriction) reduce death rates but none closes the gap to deterministic safety, with ≥12 pp residual gaps (§3.3). An exploratory router (Qwen3 LoRA, 0.6B–4B) yields flat step-level balanced accuracy (56–59%); at episode level, only 4B exceeds the always-deterministic baseline (§3.6).

Our contributions: (1) an **episode-level counterfactual diagnostic framework** that overcomes per-step oracle degeneracy (97.4% tie rate), decomposes routing signal via penalty-independent (label fraction) and penalty-dependent (DAF) metrics, and predicts routing value from a ∼50-episode pilot (validated 4/4 conditions; Appendix R); (2) **78.9% of non-tie routing labels** in ScienceWorld stem from death asymmetry, persisting across four interventions, three 7–9B model families, and 10× scaling; a step-level router recovers only task-type classification (§3.1–3.6); (3) **cross-model convergence** across three 7–9B model families (DAF within 88–90%; Llama-3.1-8B, Gemma-2-9B-it), four intervention axes, and cross-environment boundary probes (MATH, ALFWorld) confirm the framework's diagnostic utility and the robustness of the finding (§3.4–3.2).

2 Method

Our diagnostic framework quantifies what drives routing signal in interactive environments through three components: (i) episode-level counterfactual labeling that generates routing preferences from paired pure-strategy outcomes (§2.2–2.3); (ii) the Death Avoidance Factor (DAF), which decomposes the routing score gap into death-avoidance and non-death components (§2.4); and (iii) the Signal Concentration Index (SCI), which measures whether routing signal is concentrated in a few task types or distributed systematically (Appendix A). Episode-level bootstrap resampling provides 95% confidence intervals for all metrics.

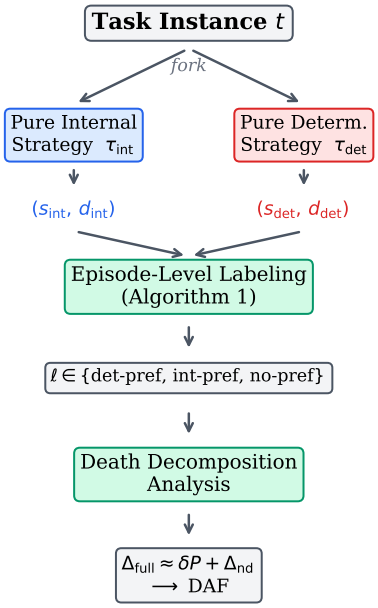


Figure 1: Counterfactual episode collection and analysis pipeline. Each task instance is run under both pure strategies independently, producing episode-level labels via Algorithm 1.

2.1 Problem Setup

Consider an interactive environment with observation space $\mathcal{O}$ and action space $\mathcal{A}$. An agent policy $\pi : \mathcal{O}^* \to \mathcal{A}$ maps observation histories to actions. We define two computation modes $\mathcal{M} = \{\texttt{int}, \texttt{det}\}$: **internal** (`int`), where the agent predicts intermediate observations via chain-of-thought reasoning (ReAct (Yao et al., 2023b)), with prediction errors compounding across steps; and **deterministic** (`det`), a privileged-information strategy that eliminates observation uncertainty—instantiated as oracle simulator (ScienceWorld), code interpreter (MATH), or admissible-action enumeration (ALFWorld); these qualitative differences affect what routing signals the framework can detect. A *routing policy* $\rho : \mathcal{O}^* \to \mathcal{M}$ selects which computation mode to use at each step; our analysis investigates what signal is available to learn ρ.

2.2 Counterfactual Episode Collection

Per-step oracle labeling assigns routing labels by comparing single-step outcomes under each computation mode. Under sparse terminal rewards, this approach degenerates: on ScienceWorld (Wang et al., 2022), 97.4% of steps produce identical intermediate scores regardless of computation mode

(5,935 of 6,094 steps across 11 prescreened task types), yielding no per-step routing signal. The 97.4% tie rate arises because ScienceWorld assigns rewards only at episode termination; intermediate steps update environment state but do not change the score, so a per-step oracle can only distinguish strategies through downstream consequences visible at episode end. Episode-level comparison is therefore the principled granularity for sparse-reward environments, not a fallback from per-step analysis: it directly measures whether one strategy produces better terminal outcomes than the other, bypassing the uninformative intermediate steps entirely.

Pure-strategy trajectories. For a task instance t and computation mode $m \in \mathcal{M}$, a *pure-strategy trajectory* is the complete episode obtained by applying mode m at every step:

$$\tau_m^{(t)} = (o_1, a_1, o_2, a_2, \ldots, o_T, a_T) \tag{1}$$

where $T \leq T_{\max}$ is the episode length (capped at $T_{\max} = 30$ steps). Both trajectories $\tau_{\texttt{int}}^{(t)}$ and $\tau_{\texttt{det}}^{(t)}$ are executed under identical initial conditions (same random seed, same task variation), so performance differences arise solely from the computation mode choice.

Each trajectory τ yields a terminal score $s(\tau) \in \mathbb{Z}$ and a death indicator $d(\tau) \in \{0, 1\}$. When the agent dies ($d = 1$), the environment imposes a penalty: $s(\tau) = -P$ where $P > 0$ ($P = 100$ in ScienceWorld). The episode pair produces a tuple:

$$\big(s^{\texttt{int}}, s^{\texttt{det}}, d^{\texttt{int}}, d^{\texttt{det}}\big) \triangleq \big(s(\tau_{\texttt{int}}),\, s(\tau_{\texttt{det}}),\, d(\tau_{\texttt{int}}),\, d(\tau_{\texttt{det}})\big) \tag{2}$$

We use Qwen2.5-7B-Instruct (Yang et al., 2024) (vLLM v0.21.0) on ScienceWorld (Wang et al., 2022), a text-based environment spanning 30 task types with a -100 death penalty. After prescreening, we collect 245 episode pairs across 11 task types (20–40 per type). The framework is applicable to any environment where paired episodes under two computation modes can be collected with binary success/failure indicators. We apply it unchanged to ALFWorld and MATH in §3.4–3.4.

2.3 Episode-Level Labeling

We define a labeling function L that maps a counterfactual pair to a routing preference:

$$L\big(\tau_{\texttt{int}}^{(t)}, \tau_{\texttt{det}}^{(t)}\big) \rightarrow \ell \in \{\texttt{det-pref}, \texttt{int-pref}, \texttt{no-pref}\} \tag{3}$$

The labeling function is *death-aware*: it prioritizes survival over score, then compares scores among same-status episodes (Algorithm 1; Appendix B). ScienceWorld scores are integer-valued, so we use exact (in)equality with no threshold. Figure 1 illustrates the overall pipeline; label distributions are reported in §3.1.

2.4 Death Avoidance Factor

To quantify the contribution of death avoidance to routing signal, we decompose the mean score gap between strategies. Let $\delta = \mathbb{E}[d^{\texttt{int}}] - \mathbb{E}[d^{\texttt{det}}]$ denote the death-rate differential and Δ_{nd} the mean score gap restricted to episodes where neither strategy dies. Partitioning episodes by death status, death-asymmetric episodes (where only one strategy dies) contribute a per-episode gap of $P + \bar{s}_{\text{surv}}$, where $\bar{s}_{\text{surv}}$ is the surviving strategy's mean score; when $\bar{s}_{\text{surv}} \ll P$ (in our data, $\bar{s}_{\text{surv}} = 0.87$, or 0.9% of P), these episodes contribute $\approx \delta P$ in aggregate. The full gap thus decomposes as:

$$\Delta_{\text{full}} \approx \underbrace{\delta \cdot P}_{\text{death contribution}} + \Delta_{\text{nd}} \tag{4}$$

The approximation residual is $29.60 - 29.37 = 0.23$ (0.8% of Δ_{full}).

We define the Death Avoidance Factor (DAF) as the fraction of the full score gap attributable to death avoidance:

$$\text{DAF} = 1 - \frac{\Delta_{\text{nd}}}{\Delta_{\text{full}}} \approx \frac{\delta P}{\delta P + \Delta_{\text{nd}}} \tag{5}$$

[1]

Since $\text{DAF} = \delta P/(\delta P + \Delta_{\text{nd}})$, it increases monotonically with P, a mathematical consequence rather than an empirical finding. For context: DAF $= 50.4\%$ at P=10, 82.6% at P=50, and 90.4% at P=100 (full curve in Appendix K). We treat the label fraction (78.9%) as the primary, penalty-invariant diagnostic and DAF as a supplementary, environment-specific measure. In ScienceWorld, P affects only post-hoc score calculation; the agent's policy (ReAct prompting) does not include P in its observation, so death rates and Δ_{nd} are P-invariant by design (Ng et al., 1999) (Appendix K).

[1] When $\Delta_{\text{nd}} < 0$ (deterministic strategy scores lower than internal on non-death episodes), DAF exceeds 1; death avoidance then overcompensates for a non-death disadvantage.

Relationship between DAF and label fraction. DAF (90.4%) exceeds the label fraction (78.9%) because death episodes carry disproportionately large score gaps (P=100 $\gg$ typical non-death $\Delta\approx$29), amplifying their weight in score-weighted aggregation. The label fraction, invariant to penalty magnitude, serves as the primary finding; DAF reflects the score-space reality under ScienceWorld's reward structure.

Statistical inference. We quantify uncertainty via episode-level bootstrap resampling (B = 10,000). Each resample draws $N = 245$ episodes with replacement, recomputes all metrics, and derives percentile-based 95% CIs. We resample at the episode level (not the step level) because the episode is the unit of counterfactual comparison. Penalty sensitivity analysis across hypothetical P values is in Appendix K.

We also measure whether death-avoidance signal is concentrated in a few task types via a Signal Concentration Index (SCI $\in [0, 1]$; Appendix A), reported in §3.1.

Scope of episode-level analysis. At step level, 97.4% of oracle comparisons yield ties (§2.2); of 2,040 training steps, only 0.3% exhibit strict directional conflict with episode labels, with 60.3% pre-event neutral (neither strategy advantaged). Episode labels thus apply correct direction to uninformative steps—a granularity mismatch, not label error—explaining why step-level router training approaches chance performance (§3.6).

3 Experiments

We analyze 11 of ScienceWorld's 30 task types (245 episodes); the 19 excluded types (63%) lack actionable routing signal due to development contamination (5), taxonomic redundancy (7), or agent-floor performance (7). Taxonomically redundant tasks share reward structure and action space, differing only in target entity (Appendix C). This scope limits generalizability: conclusions apply to the included subset, not to all ScienceWorld tasks. An anti-cherry-picking analysis confirms that death dominance holds in 18 of these 19 types, with a mean ΔDR = +38.2 pp exceeding the included tasks' +28.7 pp (Appendix C). Extended validation at n=20 on five representative excluded tasks confirms this direction ($\overline{\Delta\text{DR}}$ = +32.0 pp; Appendix S). A reasoning-depth ablation (n=120) found no significant difference between internal

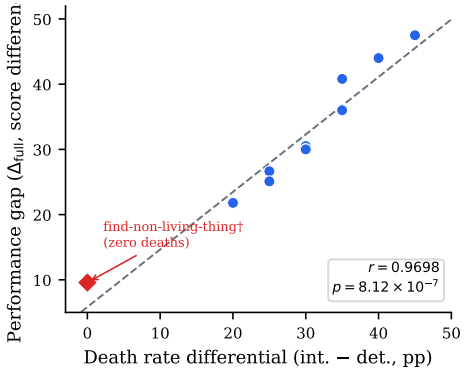


Figure 2: Internal consistency check: per-task scatter: death rate differential (internal − deterministic) vs. performance gap (Δ_{full}). $r = 0.97$ (Spearman's $\rho = 0.986$; Appendix C), $p < 10^{-6}$, $n = 11$ task types. Since $\Delta_{\text{full}} \approx \delta P + \Delta_{\text{nd}}$ with P constant, $r \to 1$ as DAF $\to 1$; the near-perfect correlation is a structural consequence of death dominance, not an independent finding (§3.1).

strategy variants ($p > 0.26$; Appendix P). Prescreening used n=5 episodes per task type; directional consistency (18/19 types favoring deterministic) rather than point estimates drives the inclusion criterion.

3.1 Death Avoidance Decomposition

Of the 95 non-tie routing labels, 75 (78.9%) arise from death asymmetry: the internal strategy triggers a catastrophic −100 penalty while the deterministic strategy survives. This penalty-independent label fraction is the primary diagnostic metric. Of these, 54 are sole-signal labels where death was the only discriminator; at P=0 these reduce to ties, but the labeling function itself is penalty-independent (Algorithm 1).

Of 245 episodes, 150 (61.2%) show no preference between strategies; routing is relevant in the remaining 95 (38.8%). Among these, death avoidance dominates. The Death Avoidance Factor, a derived penalty-dependent measure, quantifies death's contribution to the total performance gap: DAF = 90.4% at P=100 (Δ_{full} = +29.60, Δ_{nd} = +2.84; 95% CI: [81.1%, 99.0%], episode-level bootstrap, B=10,000). DAF varies monotonically with P (§2.4; Appendix K), confirming that the 78.9% label fraction provides the more robust characterization. When $\Delta_{\text{nd}} \to 0$, DAF $\to 1$ is algebraic necessity regardless of actual death dominance; conversely, when $\Delta_{\text{nd}} < 0$ (as in Reflexion $N \geq 3$; §3.3), DAF exceeds 100%, reflecting that death asymmetry more than accounts for the full gap.

Per-task correlation structure. The internal strategy dies in 42.9% of episodes versus 16.3% for deterministic (2.6×). Per-task death rate differential correlates with Δ_{full} ($r = 0.97$, $p < 10^{-6}$; Figure 2); since DAF$\to$ 1 implies $r \to 1$ by construction, the near-perfect correlation is a consistency check, not an independent finding (Appendix D). Of 245 episodes, 86 are `det_preferred`, 9 `internal_preferred`, and 150 `no_preference`.

Signal concentration and binary structure. The Signal Concentration Index (SCI = 0.2147; Appendix A) confirms low concentration: the top-3 task types account for only 39.8% of total death-rate differential. The aggregate 78.9% reflects a per-task binary structure: all 10 death-prone task types produce 100% death-asymmetry labels, while `find-non-living-thing` (zero deaths, $\Delta = +9.58$ from non-death scores) produces 0%. The framework thus diagnoses a binary property ("is this environment death-dominated?") rather than a continuous routing signal. Removing `find-non-living-thing` raises the label fraction to 100% and yields DAF ≥100% ($\Delta_{\text{nd}} \leq 0$; death avoidance fully accounts for the routing gap among remaining task types; Appendix G).

Sensitivity and seed robustness. Non-clustered episode-level bootstrap ($B = 10{,}000$) yields 95% CI $[81.1\%, 99.0\%]$ for DAF; cluster-robust CI $[73.3\%, 100.0\%]$ (K=11; Appendix E), with Cohen's $d = 3.02$ (interpret cautiously given K=11 task types). Leave-one-task-out analysis confirms stability (label fraction range: 75.9%–78.9%; Appendix G); an independent seed replication (N=220) yields DAF = 89.7%, within 0.7 pp (Appendix J). Wild cluster bootstrap (K=11, $p < 0.001$; Appendix E) and 100% directional consistency across all 28 death-prone tasks (Appendix F) provide supplementary evidence, though $K < 20$ limits cluster-level inference (Cameron et al., 2008; Webb, 2023).

3.2 Multi-Model Validation (7–9B Tier)

A Llama-3.1-8B-Instruct (Grattafiori et al., 2024) cross-model replication across all 11 task types (N=220) confirms that death dominance replicates across model families: internal death rate 41.4% vs. deterministic 11.4% (death-rate differential +30.0 pp), DAF = 88.2% (95% CI: $[80.2\%, 95.1\%]$; $\Delta_{\text{full}} = +33.17$, $\Delta_{\text{nd}} = +3.9$), with direction consistency 10/10 on death-prone tasks, 83 deterministic-preferred, 3 internal-preferred (1.4%, all death-asymmetry reversals), and 134 no-preference labels. Both models show ∼42% internal vulnerability (+26.5 vs. +30.0 pp differential; per-task details in Appendix M). A third model family, Gemma-2-9B-it (Gemma Team, 2024) (sliding window attention, vs. full attention in Qwen/Llama), independently confirms the pattern: DAF=89.7% (95% CI $[79.5\%, 98.4\%]$; N=220), with all 10 death-prone task types showing positive ΔDR (Appendix M). Three architecturally distinct models thus converge on DAF 88–90% (spread 2.2 pp); 10× scaling to 72B (§3.3) confirms persistence across capability levels (DAF=92.4%).

Condition	Death%	N
Single-trial baseline	42.9	245
Reflexion (3×)	31.4	220
Qwen2.5-72B (10×)	30.6	255
Deterministic baseline	16.3	245
10 death-prone tasks only:		
Death-prone baseline	48.0	200
Retry (5×)	43.5	200

Table 1: Internal-strategy death rates under agent improvement methods. Top: all 11 tasks (72B: independently collected at both internal and deterministic). Bottom: 10 death-prone tasks (`find-non-living-thing` excluded, 0% death at all scales). All interventions reduce death rate but none approaches deterministic-level safety.

3.3 Agent Improvement Baselines

We test whether agent-level improvement reduces death dominance (Table 1). Retry (N=5 attempts without self-correction) yields marginal reduction on death-prone tasks (43.5% vs. 48.0% baseline, $\Delta = -4.5$ pp). Reflexion (Shinn et al., 2023)[2] (verbal self-reflection between trials) reduces death to 31.4% across all 11 tasks by rescuing 29.3% of first-trial deaths, but the gap to deterministic remains 15.1 pp; on the 10 death-prone tasks, Reflexion achieves ∼35%. Under Reflexion (N=3, matched-episode analysis), $\Delta_{\text{nd}} = -0.47$, yielding DAF > 100%: the deterministic advantage is entirely attributable to death avoidance, and in non-death episodes the self-correcting agent slightly outperforms the deterministic baseline. Scaling to Qwen2.5-72B (10× parameters; both internal and

[2]Reflexion uses a fixed allocation of 20 episodes per task type (20 × 11 = 220), vs. the baseline's variable 20–40 per type (N=245).

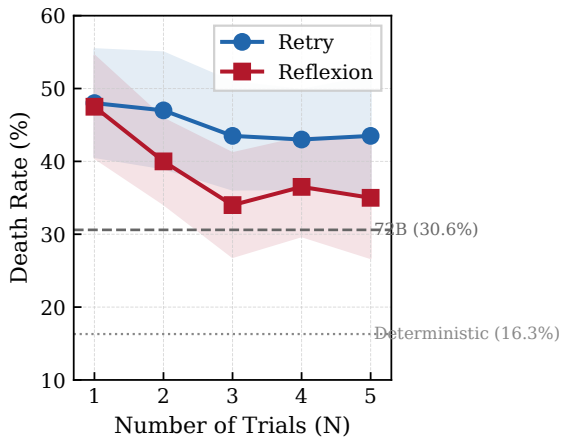


Figure 3: Death rate as a function of trial count (N) for Retry and Reflexion on 10 death-prone tasks (n=200). Retry (blue) is nearly flat; Reflexion (red) drops steeply through N=3 then shows no statistically detectable improvement beyond N=3 (Wilson CIs for N=3, 4, 5 overlap substantially). Horizontal lines: 72B scaling baseline (30.6%) and deterministic baseline (16.3%), both computed over all 11 task types; retry and Reflexion use 10 death-prone types only. Error bands: Wilson 95% CIs.

deterministic strategies independently collected at 72B, 255 episodes across all 11 task types) reduces death rate to 30.6%. The 72B deterministic strategy achieves 2.75% death rate (ΔDR = +27.8 pp), and Δ_{nd} = +2.28, yielding DAF = 92.4% (7B: 90.4%). This 10× scaling analysis (distinct from the 7–9B cross-family replication in §3.2) shows that while 72B reduces the internal death rate from 42.9% to 30.6%, DAF *increases* because the deterministic strategy also becomes safer (2.75% vs. 16.3%), making the proportional dominance of death avoidance more extreme.

Dose-response analysis. We extend both interventions to N=5 trials on death-prone tasks (n=200; Figure 3). Retry is nearly flat (48.0% → 43.5%, Δ = −4.5 pp over five trials). Reflexion drops steeply to 34.0% at N=3 (Δ = −13.5 pp, z = 2.75, p = 0.003) then stabilizes (34–36.5% for $N \geq 3$, pairwise $p > 0.5$). With n=200 paired episodes, the minimum detectable effect is ∼8 pp (α = 0.05, 80% power): the key N=1→3 decline is well-powered; the stabilization claim (N=3→5, +1.0 pp) is also supported, since a ≥8 pp improvement would be detected. We interpret the plateau as suggestive of a Reflexion ceiling (per-task breakdown in Appendix H).

Strategy	Score	95% CI	Death %
Always-Internal	−39.07	[−46.40, −32.27]	42.9
Always-Deterministic	−9.47	[−14.71, −4.29]	16.3
Random	−22.38	[−28.67, −15.75]	27.8
Oracle	−6.58	[−11.54, −1.70]	14.3
Trained Router†	−10.70	[−16.33, −5.27]	17.6

Table 2: Episode-weighted mean scores on ScienceWorld (11 task types, 20–40 episodes per type, 245 total). † Single random seed (42) per episode pair; see Appendix J for 8-seed robustness analysis. Per-task results are exploratory (n=20 episodes per task, power ≈ 50% for 20 pp effects); aggregate conclusions do not depend on individual task significance.

3.4 Cross-Environment Boundary Probes

We examine MATH and ALFWorld as boundary probes (not cross-domain validation), testing framework applicability across environments that instantiate "deterministic computation" differently (oracle simulator, code interpreter, admissible-action enumeration; see operationalization in §2.1).

MATH. On MATH Level 4–5 (Hendrycks et al., 2021) (N=795, code interpreter), internal reasoning and code interpretation achieve near-identical accuracy (73.5% vs. 73.0%, McNemar p = 0.81), with 84.4% no-preference. Without catastrophic failure, no routing signal emerges.

ALFWorld. ALFWorld (Shridhar et al., 2021) (6 task types, 135 games) provides the deterministic strategy with admissible actions, an information asymmetry rather than a computation mode switch, with no death mechanism (δ = 0, DAF = 0%). With Qwen2.5-7B, 35 of 37 non-tie labels (94.6%) favor the information-advantaged strategy (Appendix L); Llama-3.1-8B (135 games) yields 16 of 16 non-tie labels in the same direction (100%). Llama shows fewer non-tie episodes (11.9% vs. 25.9%) but perfect directional consistency, confirming that information asymmetry, not death avoidance, drives routing signal in ALFWorld regardless of model family. The framework diagnoses this qualitatively different mechanism (DAF = 0% vs. 90.4%) without modification.

3.5 Baselines

Always-deterministic (−9.47) outperforms always-internal (−39.07) by 29.60 points. The oracle improves only to −6.58, with 96.3% of decisions matching always-deterministic—an *oracle degeneracy* where an always-deterministic lookup achieves

90.5% of non-tie oracle decisions (balanced accuracy 50.0%: perfect deterministic recall, zero internal recall).

3.6 Router Training: A Preliminary Negative Result

We probe whether counterfactual labels can train a step-level router, as a diagnostic of label informativeness. The oracle margin (2.89 points) caps routing upside. The near-chance step-level accuracy admits two interpretations: (A) episode-level labels projected onto steps create ∼88% uninformative signal (60.3% pre-event neutral, 27.4% post-event), drowning learnable patterns in noise; (B) routing signal is genuinely task-type-level: there is no step-level pattern to learn. Directional conflict is only 0.3% of 2,040 training steps, but the granularity mismatch leaves minimal signal. The text augmentation ablation (25%–800% via paraphrase, no improvement) and extreme class imbalance (86/95 = 90.5% deterministic-preferred) together favor Interpretation B (non-learnability): no step-level learnable signal exists under current architectures and data regime.

An implicit router (Qwen3, LoRA; Appendix I) trained at three scales (0.6B, 1.7B, 4B) shows flat step-level balanced accuracy (56–59%). At episode level, only 4B exceeds always-deterministic ($59.0 \pm 1.9\%$, $p < 0.001$, Bonferroni-corrected across 8 comparisons); 0.6B and 1.7B remain at chance. The 4B model's marginally above-chance episode-level accuracy is consistent with task-type identification rather than step-level routing signal, as confirmed by leave-one-task-type-out analysis degrading to chance (49.2%; Appendix P). Episode-level independent evaluation yields $47.1 \pm 2.2\%$, below the majority-vote baseline of 50.0%; this independent set includes held-out task types, so the below-chance result reflects cross-task transfer failure rather than overall router incompetence. Training uses episode-level splits (Appendix I), ensuring no within-episode leakage. The high uninformative rate connects to noisy-label learning (Natarajan et al., 2013; Northcutt et al., 2021; Li et al., 2020): standard ERM on labels with ∼88% noise yields near-chance classifiers, consistent with theoretical bounds on learning from corrupted labels and empirical results in structured noise settings.

End-to-end deployment. The 4B router deployed end-to-end (N=245) scores −12.03 (95% CI: $[-17.82, -6.45]$), not exceeding always-deterministic (−9.47). Paraphrase-based augmentation (25%–800%) provides no improvement; the 3.7% internal-preferred base rate makes real-data collection expensive (Appendices J–P). A task-type-only classifier (logistic regression on one-hot task identity) achieves 64.7% balanced accuracy—higher than all step-level routers—but collapses to chance under LOTO (50.0%), matching the neural router's LOTO performance (49.2%) and confirming that routing signal is a per-task-type constant with no cross-type generalization.

4 Related Work

Tool Use and External Computation in LLMs. A growing body of work augments language models with external tools (Qu et al., 2024), spanning several functional categories. Self-supervised methods learn tool invocation from loss signals (Schick et al., 2023; Zhang et al., 2025), token-level methods embed tools in the vocabulary (Hao et al., 2023), task-level decomposition routes subtasks to specialized modules (Paranjape et al., 2023; Lu et al., 2023), and RL-based methods learn routing from reward (Feng et al., 2025). All focus on *learning to use tools*; we ask a prior question: *what signal drives the routing decision?* Concurrent work examines *when* to invoke tools: tool-call decisions correlate with uncertainty (Zeng et al., 2026), are encoded in hidden states (Sun et al., 2026), impose a "tool-use tax" (Zhang et al., 2026b), or follow necessity-utility-affordability logic (Wu et al., 2026). Our decomposition yields a concrete answer: 78.9% of non-tie routing labels trace to catastrophic-failure prevention.

Adaptive Computation and Routing. Adaptive routing traces back to learned halting policies (Graves, 2016). Mixture-of-Recursions (Bae et al., 2025) learns token-level recursion depths in the LLM era; concurrent work further diversifies routing granularity: CogRouter (Yang et al., 2026b) adapts per-step depth, ARES (Yang et al., 2026a) selects effort levels, and MTRouter (Zhang et al., 2026c) addresses multi-turn settings. RouteLLM (Ong et al., 2024) learns routers from preference data to select between strong and weak models. Concurrently, Zhang et al. (2026a) train episode-level routing policies under sparse terminal rewards, the closest setting to ours; they optimize *which* model to invoke, whereas we diagnose *what* drives the routing decision. Mixture-of-Experts architectures (Fedus et al., 2022; Lepikhin et al.,

2021) learn token-level routing across expert subnetworks; we route between computation *modes* rather than expert modules. Counterfactual data augmentation (Kaushik et al., 2020) edits inputs to flip labels; our framework executes full counterfactual *episodes* to decompose routing signal at the environment level.

Counterfactual Evaluation in RL. Prior work generates counterfactual explanations by perturbing individual actions (Olson et al., 2021; Gajcin, 2023; Amitai et al., 2024); recent and concurrent extensions target LLM agents (Pona et al., 2025; Gyevnár et al., 2025; Li et al., 2026). We instead execute complete episodes under two strategies to decompose routing signal *before* training.

LLM Agent Environments and Reasoning. ScienceWorld (Wang et al., 2022) serves as our primary testbed; ALFWorld (Shridhar et al., 2021) provides a complementary environment; AgentBench (Liu et al., 2024) offers systematic evaluation. ReAct (Yao et al., 2023b) serves as our internal baseline; Reflexion (Shinn et al., 2023) extends this with verbal self-reflection; LATS (Zhou et al., 2024) adds tree search. The concurrent SALT (Li et al., 2025) assigns step-level credit on ALFWorld, complementing our episode-level decomposition.

Safe RL and Causal Grounding. Death avoidance connects to constrained MDPs (Altman, 1999), risk-sensitive RL (Mihatsch and Neuneier, 2002), safe RL (García and Fernández, 2015; Achiam et al., 2017; Moldovan and Abbeel, 2012), and shielding (Alshiekh et al., 2018); however, classical methods assume known dynamics or differentiable constraints, unavailable for LLM agents. Ruan et al. (2024) identify catastrophic actions as a failure category; SafeRoute (Lee et al., 2025) and Agent-SafetyBench (Zhang et al., 2024) address model selection and systematic evaluation for agent safety. Our pure-strategy comparison is a simplified instance of the potential outcomes framework (Rubin, 1974), eliminating confounding without structural causal models (Pearl, 2009). Bonagiri et al. (2025) show selective quitting yields safety gains; our framework explains *why*—death avoidance dominates routing signal.

5 Discussion

Signal granularity and router failure. The router diagnostic (§3.6) reveals a structural mismatch: death avoidance is a task-type-level binary property (10/11 types carry death risk), but step-level features correlate weakly because death risk depends on multi-step trajectories. Six router variants (0.6B–4B, classification and generative) converge on near-chance step-level accuracy (56–59%); this convergent failure is consistent with signal limitation; however, extreme class imbalance (90.5% deterministic-preferred, only 9 internal-preferred episodes) makes a data-limited explanation equally plausible—distinguishing the two requires ∼1,350 paired episodes to obtain 50 minority-class samples. Cross-task-type transfer yields chance accuracy (49.2%; Appendix P). Training on pure-strategy labels but deploying under mixed-strategy routing also introduces a distribution shift (Ross et al., 2011); the negative result may partly reflect this train–deploy mismatch. Concurrent routing methods (CogRouter, ARES, MTRouter) optimize step-level signals orthogonal to death avoidance; in death-dominated environments, this task-type-level binary signal is structurally inaccessible to step features.

Non-triviality and oracle confound. While death dominance follows from the penalty structure (§2.4), two consequences were not predictable a priori: $10\times$ scaling to 72B fails to eliminate it (DAF $= 92.4\%$), and Reflexion yields DAF $> 100\%$. All four interventions leave $\geq$12 pp residual gaps (Appendix Q); cross-model DR stability (36–43%) confirms death vulnerability is a task-environment property. An oracle confound arises: is the signal merely "use the oracle"? Two observations argue otherwise: (a) $\Delta_{\text{nd}} = 2.84$ (9.6% of the total gap)—on non-death episodes, internal reasoning provides almost no advantage; (b) action-space filtering (Appendix O) reduces death rate by 19 pp but remains far above deterministic safety (16.3%). The 7 excluded agent-floor tasks, where both strategies achieve near-zero scores, represent a symmetric failure regime that the framework correctly diagnoses as "no signal."

Practical implications and actionability. The penalty-invariant label fraction (78.9%) is the primary diagnostic; DAF varies with P (Appendix K). When $f_{\text{death}} > 75\%$, always-deterministic matches learned alternatives at zero training cost (§3.5); action-space filtering reduces death rate by 19 pp (Appendix O). The framework identifies *when* always-deterministic suffices versus when router development is warranted (as in ALFWorld, where DAF=0%).

6 Conclusion

In ScienceWorld, death avoidance alone accounts for 78.9% of non-tie routing labels, replicated across three 7–9B model families (spread 2.2 pp); our episode-level counterfactual framework diagnoses this before any router is built. Cross-environment probes provide preliminary evidence that the diagnostic procedure—though not the specific operationalization of "deterministic"—transfers across environments with qualitatively different information asymmetries; a ∼50-episode pilot provides preliminary build-vs-buy signal given large effect sizes ($d = 3.02$). Future work should validate death dominance in additional catastrophic-failure environments and investigate whether lightweight death-prediction mechanisms can exploit this task-type-level binary structure without full step-level router training.

Limitations

Environmental scope. Our primary analysis uses a single environment (ScienceWorld, 11 of 30 task types) with one backbone (Qwen2.5-7B-Instruct), though extended validation of representative excluded tasks confirms the pattern (Appendix S). The -100 death penalty amplifies DAF; environments with softer penalties would exhibit lower DAF values (Appendix K). The 19 excluded task types (63%) lack routing signal because both strategies typically time out within 30 steps; increasing the step budget could surface non-death-dominated signal that our current setup does not capture. Cross-environment analyses on MATH and ALFWorld (§3.4, §5) provide preliminary supporting evidence but do not conclusively establish generalization. A Llama-3.1-8B-Instruct cross-model replication (11 task types, N=220; §3.2) shows death dominance replicates with direction consistency 10/10 while vulnerability magnitude varies (+30.0 pp vs. Qwen +26.5 pp), though both models are in the 7–8B parameter range; Qwen2.5-72B (10× scale) reduces death rate but does not eliminate death dominance (§3.3); validation beyond 72B remains future work. The 72B probe is limited to a single model without cross-model replication at scale; stronger models may develop non-death strategies that reduce death dominance. Our quantitative results (DAF = 90.4%, label fractions) are specific to ScienceWorld; the framework's generality claim is methodological (the diagnostic procedure applies to any environment with asymmetric fatal outcomes), not empirical (the specific percentages may differ across environments).

Methodological constraints. Our internal-reasoning baseline uses single-pass ReAct (Yao et al., 2023b). Reflexion (Shinn et al., 2023) reduces death rate from 42.9% to 31.4% (§3.3), but more aggressive self-correction or Tree-of-Thought backtracking (Yao et al., 2023a) may further reduce internal death rates. Episode-level labels assign a single routing preference to entire episodes, which may be suboptimal for individual steps (particularly those preceding death events). However, of 2,040 training steps only 0.3% exhibit strict directional conflict (step-level oracle opposite to episode label); the remaining apparent mislabeling consists of episode labels applied to steps where neither strategy has a step-level advantage (60.3% pre-event neutral). The issue is granularity mismatch (applying correct-direction labels to uninformative steps), not directional error. Training on pure-strategy trajectories also introduces distribution shift at deployment (Ross et al., 2011). Under mixed-strategy routing with oracle death prediction, switching to deterministic at the step before death could reduce the internal death rate; even under optimistic assumptions (perfect step-level death prediction), a residual gap of >10 pp between strategies is expected, as fatal trajectories involve cumulative hallucination errors preceding the death event. More broadly, projecting episode-level labels onto 10–50 sequential steps with path-dependent state introduces compounding error: early routing decisions alter the state trajectory, potentially invalidating later labels derived from the episode outcome.

Statistical limitations. Episodes within the same task type are not fully independent because they share environment dynamics, though they use different random seeds. Cluster bootstrap with K=11 task-type clusters yields wider CIs ($[73.3\%, 100.0\%]$); with K=11 falling below the $K \geq$ 20–50 threshold recommended by Cameron et al. (2008); Webb (2023), readers should rely on the non-clustered bootstrap CI ($[81.1\%, 99.0\%]$). A wild cluster bootstrap (K=11, $p < 0.001$; Appendix E) and 100% directional consistency across all 28 death-prone tasks (Appendix F) mitigate this concern.

Pure-strategy upper bound. Our counterfactual analysis compares pure strategies (always-internal

vs. always-deterministic). In mixed routing, the router can selectively invoke deterministic mode on high-risk steps, potentially reducing death asymmetry. Thus, DAF=90.4% represents an upper bound on death avoidance's contribution to optimized routing; the actual contribution under mixed strategies may be lower.

Diagnostic scope. Our framework is diagnostic rather than prescriptive: it identifies what drives routing decisions but does not optimize the routing policy itself. The framework assumes binary catastrophic failure; environments with graded penalties would require modifying Algorithm 1's death-priority rule to a continuous severity weighting. Extending the counterfactual decomposition to prescribe optimal mixed-strategy routing (for example, by combining DAF analysis with online policy learning) is a natural next step.

Ethics Statement

This work uses ScienceWorld (Wang et al., 2022), a publicly available text-based simulation environment, and does not involve human subjects or private data. All evaluation environments (ScienceWorld, ALFWorld, MATH) are released under permissive open-source licenses (Apache 2.0 / MIT). All models used (Qwen2.5 family, Llama-3.1-8B, Gemma-2-9B-it) are publicly available for research use. All language models are accessed via standard APIs or publicly released weights. Experiments were conducted on 2× NVIDIA RTX 5090 (32GB) and 4× NVIDIA RTX 6000 PRO (96GB), totaling approximately 200 GPU-hours, comprising ∼140 hours for episode collection (counterfactual inference across three 7–9B model families plus Qwen2.5-72B backbone scaling, Reflexion and retry baselines, and cross-environment validation on ALFWorld and MATH), ∼35 hours for router training (LoRA fine-tuning of Qwen3-0.6B/1.7B/4B across 4–8 seeds per scale, including augmentation and ablation variants), and ∼25 hours for evaluation and analysis. Code and data will be released upon acceptance.

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Appendix Overview. A. Signal Concentration Index · B. Labeling Quality Audit · C. Per-Task Decomposition · D. Correlation Robustness · E. Cluster Bootstrap Sensitivity · F. Per-Task CIs · G. Leave-One-Task-Out DAF · H. Dose-Response Breakdown · I. Router Training Details · J. Seed Robustness · K. Penalty Sensitivity · L. Cross-Environment Details · M. Cross-Model Verification · N. Failure Mode Analysis · O. Grounding Ablation · P. Ablation Studies · Q. Threshold Routing · R. Decision Framework · S. Extended Validation · T. Agent Prompt Templates

Algorithm 1 Episode-Level Counterfactual Labeling

Require: Task instance t, computation modes $\mathcal{M} = \{\texttt{int}, \texttt{det}\}$
Ensure: Label $\ell \in \{\texttt{det-pref}, \texttt{int-pref}, \texttt{no-pref}\}$

1: **for** each $m \in \mathcal{M}$ **do**
2: Run episode τ_m with pure-strategy m
3: Record score s_m and death indicator d_m
4: **end for**
5: **if** $d_{\texttt{int}} = 1$ **and** $d_{\texttt{det}} = 0$ **then**
6: $\ell \leftarrow$ `det-pref` {death asymmetry}
7: **else if** $d_{\texttt{int}} = 0$ **and** $d_{\texttt{det}} = 1$ **then**
8: $\ell \leftarrow$ `int-pref` {death asymmetry}
9: **else if** $d_{\texttt{int}} = d_{\texttt{det}} = 1$ **then**
10: $\ell \leftarrow$ `no-pref` {both dead}
11: **else if** $s_{\texttt{det}} > s_{\texttt{int}}$ **then**
12: $\ell \leftarrow$ `det-pref` {score comparison}
13: **else if** $s_{\texttt{int}} > s_{\texttt{det}}$ **then**
14: $\ell \leftarrow$ `int-pref` {score comparison}
15: **else**
16: $\ell \leftarrow$ `no-pref` {tied alive}
17: **end if**
18: **return** ℓ

Assumption: The death-first priority (lines 5–10) is valid when P exceeds the maximum partial score, guaranteeing that a dead episode never outscores a surviving one. ScienceWorld satisfies this (P=100, max partial score =100).

A Signal Concentration Index

Let K denote the number of task types and c_k the contribution of task type k to the aggregate routing signal, defined as the fraction of total death-rate differential attributable to task k:

$$c_k = \frac{n_k \cdot \delta_k}{\sum_{j=1}^{K} n_j \cdot \delta_j} \tag{6}$$

where n_k is the number of episodes and $\delta_k = d_k^{\texttt{int}} - d_k^{\texttt{det}}$ is the per-task death-rate differential. The SCI is the Gini coefficient over $\{c_1, \ldots, c_K\}$:

$$\text{SCI} = \frac{\sum_{i=1}^{K} \sum_{j=1}^{K} |c_i - c_j|}{2K \sum_{k=1}^{K} c_k} \tag{7}$$

B Labeling Quality Audit

Note that episode-level labels are used to evaluate step-level predictions, introducing a granularity mismatch rather than independent validation. A potential concern is oracle circularity: whether episode-level labels reflect tie-breaking artifacts rather than genuine computation mode differences. We audit all 245 labels. Of the 150 `no_preference` labels (61.2%), 115 (46.9%) are both-alive ties and 35 (14.3%) are both-dead, all excluded from router training. The remaining 95 training labels are entirely genuine: 75 from death asymmetry (70 deterministic-preferred, 5 internal-preferred) and 20 from score comparison with both strategies surviving (16 deterministic-preferred, 4 internal-preferred). Zero labels depend on tie-breaking. Retraining the router using only death-asymmetry labels yields step-level balanced accuracy of 57.3%, within 1σ of the full-label retrain (57.6 $\pm$ 0.5%, step-level Trainer evaluation); score-comparison labels contribute negligible additional signal. (Step-level metrics are used here for the relative comparison; as shown in §3.6, absolute generalization to new episodes is near chance.) The 97.4% step-level tie rate (§2.2) further motivates episode-level over step-level labeling.

C Per-Task Decomposition

We use ScienceWorld v1.2.3 (Wang et al., 2022). We select 11 of ScienceWorld's 30 task types for analysis. Types are excluded when they were used for pipeline development (5 tasks, standard hold-out practice), are taxonomically redundant with a retained family representative (7 tasks), or exhibit near-zero task progress under both strategies (7 tasks, agent capability floor).[3]

Task selection and anti-cherry-picking. Eighteen of the 19 excluded task types (95%) exhibit death-asymmetric patterns (Table 4): internal-reasoning death rates range from 40% to 60% (mean 53.9%), while deterministic death rates range from 0% to 40% (mean 15.8%), yielding a mean death-rate differential of +38.2 pp, exceeding the +28.7 pp for included tasks. The included subset is thus *less* death-asymmetric than the excluded subset, the opposite of what a cherry-picking hypothesis would predict. The sole exception is `grow-plant`, an intrinsically safe task with zero deaths under both strategies, paralleling `find-non-living-thing` in the included set. The 19 excluded tasks fall into three independently motivated categories: five used during pipeline development (standard hold-out practice), seven

[3] Excluded tasks exhibit even stronger death dominance than included tasks (ΔDR = +38.2 pp vs. +28.7 pp; Table 4).

Task	n	Δ_{full}	D%$_{\text{int}}$	D%$_{\text{det}}$	D / I / N
id-life-stages-1	20	+47.50	55	10	9 / 0 / 11
lifespan-long	25	+44.00	56	16	11 / 1 / 13
lifespan-l-then-s	20	+40.80	45	10	8 / 1 / 11
inclined-plane	20	+36.00	55	20	7 / 0 / 13
freeze	20	+30.50	55	25	6 / 0 / 14
meas-melt-point	20	+30.05	55	25	6 / 0 / 14
id-life-stages-2	20	+30.00	50	20	6 / 0 / 14
find-animal	20	+26.65	50	25	5 / 0 / 15
grow-fruit	20	+25.10	45	20	6 / 1 / 13
chem-mix	20	+21.80	45	25	6 / 2 / 12
find-non-living-thing†	40	+9.58	0	0	16 / 4 / 20
All (11 tasks)	**245**	**+29.60**	**42.9**	**16.3**	**86 / 9 / 150**

Table 3: Per-task decomposition of the death avoidance effect. D%: death rate for internal (int) and deterministic (det) strategies. D/I/N: episode labels (det-preferred / internal-preferred / no-preference). Sorted by Δ_{full} descending. The *All* row is an episode-weighted aggregate across 245 episodes. † Only task with zero deaths under both strategies; n=40 because data was collected in two batches, with the second batch adding 20 episodes to increase statistical power for this zero-death outlier.

Task	Grp	n	D%$_{\text{int}}$	D%$_{\text{det}}$	ΔDR	Excl. reason
boil	A	20	50	20	+30	dev set
change-state-of-matter	A	20	55	15	+40	dev set
chem-mix-paint-secondary	A	20	45	10	+35	dev set
grow-plant	A	20	0	0	0	dev set
melt	A	20	55	15	+40	dev set
chem-mix-paint-tertiary	B	5	40	20	+20	family redundancy
find-living-thing	B	5	60	20	+40	family redundancy
find-plant	B	5	60	0	+60	family redundancy
inclined-friction-named	B	5	60	20	+40	family redundancy
inclined-friction-unnamed	B	5	60	0	+60	family redundancy
lifespan-shortest	B	5	60	40	+20	family redundancy
measure-melt-known	B	5	60	0	+60	family redundancy
mendelian-known	C	5	60	20	+40	agent floor
mendelian-unknown	C	5	60	40	+20	agent floor
power-component	C	5	60	20	+40	agent floor
power-renewable	C	5	60	0	+60	agent floor
test-conductivity	C	5	60	20	+40	agent floor
test-conductivity-unknown	C	5	60	20	+40	agent floor
use-thermometer	C	5	60	20	+40	agent floor
Mean (19 excluded)			**53.9**	**15.8**	**+38.2**	
cf. included (11)			*46.5*	*17.8*	*+28.7*	

Table 4: Death-rate analysis of all 19 excluded task types. D%: death rate (score ≤ -100). ΔDR = D%$_{\text{int}}$ − D%$_{\text{det}}$. 18 of 19 excluded tasks (95%) exhibit positive death asymmetry (ΔDR $\geq +20$ pp). Exclusion groups: A = development set (excluded to prevent circular evaluation), B = taxonomically redundant with an included family representative, C = agent-floor tasks (backbone achieves near-zero progress regardless of strategy). The mean ΔDR among excluded tasks (+38.2 pp) exceeds that of included tasks (+28.7 pp), confirming that task selection does not bias toward death-prone tasks. Group B–C estimates are based on n=5 prescreening episodes per task and should be interpreted as directional evidence rather than precise rate estimates.

taxonomically redundant with included family representatives (deduplication), and seven where the 7B backbone achieves near-zero task progress regardless of strategy (agent capability floor). In no case was a task excluded because death dominance was absent. Exclusion is driven by absence of non-death routing signal ($\Delta_{\text{nd}} = 0$ for 53% of excluded tasks) and standard deduplication practices. `find-non-living-thing` serves as an internal counterexample: zero deaths under both strategies yet included because it exhibits non-death routing signal ($\Delta_{\text{full}} = +9.58$ entirely from score differences), and deterministic is not always preferred (4 of 20 non-tie labels favor internal).

D Correlation Robustness

The Pearson $r = 0.9698$ reported in §3.1 is robust across multiple diagnostics. Spearman rank correlation is $\rho = 0.986$ ($p = 2.4 \times 10^{-8}$); the relationship is not driven by outliers or non-linearity. A nonparametric bootstrap (10,000 resamples) yields 95% CI $[0.952, 0.995]$.

E Cluster Bootstrap Sensitivity Analysis

With K=11 task-type clusters, cluster bootstrap falls well below the $K \geq$20–50 recommended for reliable inference (Cameron et al., 2008; Webb, 2023). The narrower CIs obtained reflect known small-sample bias toward underestimation of variance, not tighter inference.

Episodes within the same task type share environment dynamics, raising the concern that the effective sample size is closer to K (task types) than N (episodes). We address this with cluster-robust inference treating each task type as a cluster (K=11, all 11 included task types, 245 episodes). All methods confirm significance with $p < 0.01$, and the 95% CI $[0.73, 1.00]$ excludes zero by a wide margin. With K=11 task types, ICC estimation itself carries substantial uncertainty; the high within-task consistency of death patterns (all-or-nothing per task type) suggests high ICC, which explains the CI narrowing under cluster resampling.

We retain this analysis for transparency; readers should rely on the non-clustered episode-level bootstrap (§3.1) as the primary inference.

F Per-Task Death Rate Confidence Intervals

We compute bootstrap 95% CIs for the death-rate differential (ΔDR = DR$_{\text{int}}$ − DR$_{\text{det}}$) for all 30 ScienceWorld task types. Of 30 tasks, 28 are death-prone (at least one death under either strategy); the remaining 2 (`grow-plant`,

Task removed	r_{LOO}
id-life-stages-1	0.963
lifespan-long	0.969
lifespan-l-then-s	0.976
inclined-plane	0.969
freeze	0.971
meas-melt-point	0.972
id-life-stages-2	0.972
find-animal	0.970
grow-fruit	0.973
chem-mix	0.969
find-non-living-thing	0.980

Table 5: Leave-one-out Pearson r for the death rate differential vs. Δ_{full} correlation. Full $r = 0.970$; LOO range $[0.963, 0.980]$. No single task type is a leverage point.

Method	p-value	95% CI
Cluster-robust t-test (t=10.0, df=10)	< 0.001	$[0.73, 1.00]$
Wild cluster bootstrap (B=9,999)	< 0.001	$[0.73, 1.00]$
Exact permutation (2,048 perms)	0.002	—
Sign test	< 0.001	—

Table 6: Cluster-level tests of death-rate asymmetry (K=11 task clusters, 245 episodes). 10 of 11 clusters show internal DR > deterministic DR; the sole exception (`find-non-living-thing`) has zero deaths under both strategies. Cohen's $d = 3.02$. The exact permutation $p = 0.002$ ($4/2{,}048$) reflects the near-unanimous directional consistency.

`find-non-living-thing`) have zero deaths under both strategies.

Across all 28 death-prone tasks, internal DR exceeds deterministic DR without exception (100% directional consistency). The pooled death-rate differential is $+37.3\%$ (95% CI: $[29.7\%, 44.4\%]$). Fifteen of 28 tasks are individually significant (CI excludes zero); the 13 non-significant tasks all have n=5 prescreening episodes, where statistical power is insufficient to detect even large effects (e.g., a task with $\Delta\text{DR} = +40$ pp at n=5 has a CI of ± 43 pp). The universal directional consistency, combined with the pooled significance, confirms that death asymmetry is not driven by a subset of outlier tasks.

G Leave-One-Task-Out DAF Sensitivity

Table 7: Leave-one-task-out DAF sensitivity (penalty-independent label fraction). Among the 10 death-prone tasks, removing any single task changes DAF by at most 3.0 pp (range: 75.9%–78.9%). Removing *Find Non-Living Thing*, the sole death-free task that contributes all 20 score-comparison episodes, raises DAF to 100% by construction.

Removed Task	N_{rem}	DAF (%)	Δ (pp)
(none, full)	245	78.9	—
Find Non-Living[†]	205	100.0	+21.1
Lifespan Longest	220	75.9	−3.0
Life Stages 1	225	76.7	−2.2
Lifespan Long+Short	225	76.7	−2.2
Chemistry Mix	225	77.0	−1.9
Grow Fruit	225	77.3	−1.7
Inclined Plane Angle	225	77.3	−1.7
Freeze	225	77.5	−1.4
Life Stages 2	225	77.5	−1.4
Melt Pt Unknown	225	77.5	−1.4
Find Animal	225	77.8	−1.2

[†]Only death-free task; contributes all score-comparison labels. Sorted by $|\Delta|$ descending.

H Per-Task Dose-Response Breakdown

Table 8 reports per-task death rates for Reflexion at N=3, 4, and 5 on the 10 death-prone tasks (n=20 episodes per task). Per-task variance is high (up to 25 pp range for `measure-melting-point`), but the aggregate smooths to a 2.5 pp band, confirming that the stabilization conclusion is robust at the aggregate level despite task-level noise. `freeze` is consistently the hardest task (50–60% DR across all N).

Table 8: Per-task death rates (%) for Reflexion at N=3, 4, 5 on 10 death-prone tasks (n=20 per task). Aggregate death rates confirm stabilization at $N \geq 3$ (range: 34.0–36.5%).

Task	N=3	N=4	N=5
chemistry-mix	25.0	35.0	20.0
find-animal	45.0	30.0	45.0
freeze	50.0	50.0	60.0
grow-fruit	30.0	25.0	25.0
identify-life-stages-1	35.0	35.0	35.0
identify-life-stages-2	25.0	25.0	30.0
inclined-plane-angle	20.0	30.0	35.0
lifespan-longest	45.0	40.0	40.0
lifespan-long+short	30.0	45.0	35.0
measure-melting-pt	35.0	50.0	25.0
Aggregate	**34.0**	**36.5**	**35.0**

I Router Training Details

We use Qwen3-1.7B for the router to avoid circular dependency with the Qwen2.5-7B data-generating model.

With gradient accumulation of 1 (effective batch size 4), routing performance is highly sensitive to the evaluation split; this suggests overfitting to specific train/validation partition boundaries. With gradient accumulation of 8 (effective batch size

32), step-level performance stabilizes across splits ($\sim$54% balanced accuracy). All reported results use gradient accumulation 8.

Under step-level evaluation with a fixed data split, cross-seed balanced accuracy ranges from 52.7% to 58.1%, but internal recall varies from 51.5% to 81.8%, which reflects threshold sensitivity. However, episode-level evaluation reveals that this apparent discrimination does not generalize: balanced accuracy drops to $47.1 \pm 2.2\%$ across 8 cross-split seeds (Appendix J). Deployed end-to-end, the trained router allocates 97.0% of episodes to deterministic computation with 17.6% death rate, and misrouting cost (death from internally-routed episodes) exceeds the benefit of correctly routing rare internal-preferred cases.

Evaluation methods. All results use *generation-based evaluation*: the model generates a routing label as text (`enable_thinking=False`), parsed for classification. We evaluate at two granularities: *step-level* (steps from the same episode may appear in both train and validation sets) and *episode-level* (entire episodes held out). Step-level accuracy ($54.1 \pm 5.4\%$) exceeds chance (the router captures some signal), but episode-level accuracy ($47.1 \pm 2.2\%$, below chance) shows this does not generalize to unseen episodes. Paraphrase-based text augmentation (Appendix J) provides no improvement; real-data scaling remains open. Always-deterministic is trivially near-optimal given DAF $= 90.4\%$. The main text (§3.6) reports controlled 4-seed results at three scales; the model-size ablation (Appendix P) provides further detail.

Hyperparameter	Classification Head	Implicit Router
Base model	Qwen3-1.7B	Qwen3-1.7B
Architecture	Linear + LoRA	SFT + LoRA
LoRA r / α	16 / 32	16 / 32
Learning rate	2e-5	2e-4
Optimizer	AdamW	AdamW
Epochs	3	3
Batch size (per device)	4	4
Gradient accumulation	8	8
Warmup ratio	0.1	0.1
Weight decay	0.01	0.01
Max sequence length	512	512
Seeds	{42}	{42, 123, 456, 789}

Table 9: Router training hyperparameters. Both architectures use the same base model with LoRA adaptation. The classification head degenerates across all configurations; the implicit router results are reported as 4-seed mean±std.

Agent inference parameters. For ScienceWorld, each agent step uses temperature T=0.3 and maximum generation length of 256 tokens, served via vLLM v0.21.0. Top-p is left at the vLLM default of 1.0 (no nucleus sampling). For MATH, the agent uses greedy decoding (T=0.0) with a maximum of 2,048 tokens per generation. For ALFWorld, we use T=0.3 with 256 maximum tokens. All three environments use Qwen2.5-7B-Instruct (Yang et al., 2024) as the backbone.

J Seed Robustness

We assess router generalization at two evaluation granularities: *step-level*, where steps from the same episode may appear in both train and validation sets, and *episode-level*, where entire episodes are held out. Both use cross-split evaluation (seeds $\{42, \ldots, 49\}$, 8 seeds total), where the data partition and training seed both vary. Table 10 reports per-seed results.

Eval. Granularity	Seed	Bal. Acc. (%)
Step-level (same-episode leakage)	42	57.06
	43	63.99
	44	55.43
	45	56.14
	46	55.45
	47	50.08
	48	48.03
	49	59.30
	Mean±std	54.1 ± 5.4
Episode-level (held-out episodes)	42	48.1
	43	47.9
	44	49.6
	45	49.4
	46	48.5
	47	43.1
	48	45.4
	49	44.8
	Mean±std	47.1 ± 2.2
Always-det (= majority-vote)		50.0

Table 10: Per-seed router performance under cross-split evaluation (8 seeds, generation-based, `enable_thinking=False`). Step-level evaluation allows steps from the same episode in both train and validation sets (same-episode leakage). Episode-level evaluation holds out entire episodes; the resulting $47.1 \pm 2.2\%$ is below the 50% chance baseline. A text augmentation ablation (Table 11) confirms this failure persists under the current data distribution and augmentation regime.

The contrast between step-level ($54.1 \pm 5.4\%$) and episode-level ($47.1 \pm 2.2\%$) evaluation reveals a generalization gap: the step-level signal likely reflects memorization of episode-specific context (enabled by same-episode leakage) rather than extraction of generalizable routing features. At episode

level, the router performs below chance across all 8 seeds (range: 43.1%–49.6%). Paraphrase-based text augmentation (Table 11) from 25% to 800% (32× range) provides no improvement, converging to the always-deterministic baseline (50%). Paraphrase augmentation inflates the majority class without generating novel routing patterns; genuine episode-level scaling (collecting more counterfactual pairs) remains an open question. With DAF $= 90.4\%$ and all task types deterministic-majority (86:9 label ratio), always-deterministic trivially solves the episode-level routing problem, leaving the step-level router with no exploitable signal under current architectures.

Text augmentation ablation. To test whether the episode-level generalization failure stems from data scarcity, we vary training set size from 25% to 800% of the original via paraphrase-based text augmentation (synonym substitution + template variation on step observations), holding the episode-level validation set fixed. This tests augmentation-based scaling, not genuine episode-level scaling (which would require new counterfactual collections). Table 11 reports the results.

Scale	Samples	Det%	Ep. Bal. Acc. (%)
25%	433	∼90	49.5 ± 0.3
50%	867	∼90	51.5
100%	1,735	89.6	49.9 ± 0.1
200%	3,470	94.2	50.0 ± 0.0
800%	13,880	∼97.7	51.0 ± 1.1
Always-det	—	—	50.0

Table 11: Text augmentation ablation: episode-level balanced accuracy across training set sizes (25%–800% of original via paraphrase, 32× range). Internal recall $= 0.0$ at all scales (complete majority-class collapse). Det%: fraction of deterministic-preferred labels in training data. The 50% scale is a single seed; all others report mean±std across seeds.

All scales hover at ∼50% episode-level balanced accuracy (equivalent to the always-deterministic baseline), with no monotonic improvement as data increases. Augmentation-based scaling provides no improvement at any scale; however, episode-level minority-class diversity remains a potential confound that cannot be resolved without collecting more internal-preferred episodes. The ground-truth internal-preferred rate (9/245 $= 3.7\%$) means collecting 50 internal-preferred episodes would require approximately 1,350 paired episodes at current environment dynamics, a $5.5\times$ expansion with diminishing return on a 9.6% signal. The failure points to a limitation of the current data distribution: when DAF $= 90.4\%$ and all task types are deterministic-majority (86:9 label ratio), the episode-level routing problem is trivially solved by always choosing deterministic, and the step-level router finds no exploitable signal under the current architecture and data regime.

Augmentation disproportionately increases the majority (deterministic) class: Det% rises from 89.6% at 100% to 94.2% at 200% and ∼97.7% at 800%. However, the complete collapse to majority-class prediction (internal recall $= 0.0$) occurs even at the smallest unaugmented scale (25%, ∼90% det); this is not an augmentation artifact. Balanced accuracy corrects for class imbalance by averaging per-class recall, so the metric itself is not biased by the skew.

Decomposition metrics are stable under bootstrap resampling (B=10,000 resamples of the 245-episode dataset): DAF $= 90.4\% \pm 4.2\%$ (95% CI $[81.1\%, 99.0\%]$); Pearson $r = 0.975 \pm 0.011$ (95% CI $[0.952, 0.995]$). The death avoidance dominance finding is robust to both resampling and split perturbation.

Per-task-type evaluation breakdown. To test whether the router learns transferable routing strategies or merely task-type classification, we evaluate existing checkpoints (8 seeds) separately on each of the 11 task types. Of the 11 task types, 7 have *all-deterministic* labels (no internal-preferred episodes exist), making balanced accuracy undefined; the router has nothing to learn beyond the trivial always-deterministic default. For the remaining 4 task types where both classes are present, balanced accuracy ranges from 50.0% to 73.0%, but this apparent signal does not indicate cross-task-type routing transfer: the router is effectively classifying task types (which correlate with death risk) rather than extracting a generalizable routing strategy. This complements the text augmentation ablation: routing signal does not transfer across task types because the signal *is* the task type, i.e., whether a given task carries death risk. The 4B capacity scaling result (Appendix P) suggests capacity is one factor, but even 4B's 59.0% episode-level accuracy does not demonstrate cross-task-type transfer.

P	DAF(P)	95% CI
0	7.4%	[−39.4, 82.5]
1	14.8%	[−28.6, 89.6]
5	35.4%	[−1.8, 92.7]
10	50.4%	[18.5, 93.3]
25	70.7%	[48.0, 96.3]
50	82.6%	[67.2, 98.2]
100*	90.4%	[81.1, 99.0]
200	94.9%	[89.9, 99.5]
500	97.9%	[95.7, 99.8]

Table 12: Sensitivity of DAF to death penalty magnitude P. $\text{DAF}(P) = 1 - \Delta_{\text{nd}}/\Delta_{\text{full}}(P)$, where $\Delta_{\text{full}}(P) \approx \delta \cdot P + \Delta_{\text{nd}}$, $\delta = 0.2653$ is the death-rate differential, and $\Delta_{\text{nd}} = 2.84$ (strict neither-died filtering). Recomputed from 245 episode-level data; bootstrap 95% CIs (B=10,000). *Default setting.

K Penalty Sensitivity Analysis

The non-death score gap $\Delta_{\text{nd}} = 2.84$ is invariant across penalty levels; it depends only on surviving-episode performance, not on P. As P increases, Δ_{full} grows (16.33 at P=50, 29.60 at P=100, 56.13 at P=200) while Δ_{nd} remains fixed, mechanically increasing DAF.

Mechanism boundary: death specificity. The $r = 0.97$ correlation between per-task death-rate differential and Δ_{full} is specific to catastrophic death (score $= -100$). Broader failure definitions dissolve the correlation entirely: defining "failure" as any non-positive score (T=0) yields $r = -0.13$ ($p = 0.70$), because non-death episodes show near-zero performance differential between strategies ($\Delta_{\text{nd}} = 2.84$). Table 13 quantifies this specificity across penalty magnitudes.

P	r	p	Sig. (α=0.05)
1	−0.44	0.178	No
5	−0.29	0.382	No
10	−0.08	0.813	No
25	0.51	0.108	No
50	0.86	0.0006	Yes
100*	0.97	8×10^{-7}	Yes
200	0.99	1×10^{-9}	Yes

Table 13: Pearson r between per-task death-rate differential and $\Delta_{\text{full}}(P)$ across penalty magnitudes. The correlation is non-significant at low penalties and emerges at $P \geq 50$, which reflects the transition from non-death-dominated to death-dominated score gaps. *Default ScienceWorld setting.

This sharpens our central finding: death is not merely the dominant component of routing signal; it is effectively the *only* component that generates differential routing signal in ScienceWorld. At low penalties, the death contribution is small and Δ_{full} reduces to $\Delta_{\text{nd}} \approx 2.84$, which is near-uniform across task types (mean non-death gap $= -1.19$ excluding the `find-non-living-thing` outlier); consequently, death-rate differential has no predictive power. As P increases, the death term dominates Δ_{full}, and the correlation sharpens because death rates *are* the performance gap. The DAF values in Table 12 increase monotonically with P, but this is a mathematical consequence of the decomposition ($\text{DAF} = \delta P/(\delta P + \Delta_{\text{nd}})$), not evidence of mechanism robustness.

L Cross-Environment Details

Figure 4 and Table 14 summarize the label distributions across all three environments.

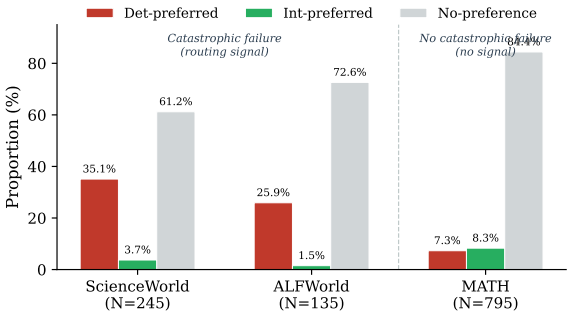


Figure 4: Cross-environment label distributions. ScienceWorld and ALFWorld both show deterministic preference, but through different mechanisms: death avoidance (ScienceWorld) vs. information asymmetry from admissible-action access (ALFWorld). MATH shows balanced labels, confirming no routing signal without catastrophic failure.

Environment	N	Det%	Int%	NoPref%	Death?
ScienceWorld	245	35.1	3.7	61.2	Yes (−100)
MATH	795	7.3	8.3	84.4	No
ALFWorld	135	25.9	1.5	72.6	No†

Table 14: Cross-environment label distribution. "Deterministic" denotes qualitatively different computation modes: ScienceWorld = oracle simulator with ground-truth state; MATH = Python code interpreter; ALFWorld = admissible-action enumeration. † ALFWorld's deterministic advantage stems from information asymmetry, not death avoidance.

Table 15 provides the per-task-type breakdown for ALFWorld. Unlike ScienceWorld, ALFWorld has no death mechanism ($\delta = 0$, DAF $= 0\%$); the entire performance gap Δ_{full} reflects information asymmetry from admissible-action access.

`pick_and_place` shows the strongest effect ($\Delta = 0.563$, 18 of 32 episodes det-preferred), while `pick_two_obj_and_place` shows the weakest ($\Delta = 0.087$), consistent with multi-object tasks requiring more sequential reasoning that admissible actions alone cannot resolve. Only `pick_heat_then_place` has a 95% CI crossing zero (n=16).

Task Type	n	Succ%$_{\text{int}}$	Succ%$_{\text{det}}$	D / I / N	Δ_{full}
pick_and_place	32	0.0	56.3	18 / 0 / 14	+0.563
look_at_obj_in_light	13	15.4	38.5	3 / 0 / 10	+0.231
pick_cool_then_place	24	0.0	16.7	4 / 0 / 20	+0.167
pick_clean_then_place	27	3.7	18.5	5 / 1 / 21	+0.148
pick_heat_then_place	16	6.3	18.8	3 / 1 / 12	+0.125
pick_two_obj_and_place	23	0.0	8.7	2 / 0 / 21	+0.087
All (6 types)	**135**	**3.0**	**27.4**	**35 / 2 / 98**	**+0.244**

Table 15: Per-task-type decomposition for ALFWorld (135 games, Qwen2.5-7B-Instruct). D/I/N: deterministic-preferred / internal-preferred / no-preference. No death mechanism exists; Δ_{full} is driven entirely by information asymmetry from admissible-action access. Sorted by Δ_{full} descending.

MATH subset composition. The N=795 MATH problems span two categories from the MATH test set (Hendrycks et al., 2021) (Levels 4–5): Algebra (585, 73.6%) and Counting & Probability (210, 26.4%). The category imbalance reflects the relative sizes in the original test set.

M Cross-Model Verification

We replicate the counterfactual analysis with Llama-3.1-8B-Instruct across all 11 task types (N=220 episode pairs, 20 per type; Table 16). The internal strategy dies in 41.4% of episodes versus 11.4% for deterministic (death-rate differential +30.0 pp), with 83 deterministic-preferred, 3 internal-preferred (1.4%, all death-asymmetry reversals), and 134 no-preference labels. Both models exhibit nearly identical internal vulnerability (∼42%: Qwen 42.9%, Llama 41.4%), confirming that death-prone failure is empirically dominated by the task environment rather than model-specific factors. The differential magnitude varies (+26.5 pp Qwen, +30.0 pp Llama) because deterministic strategy safety is model-dependent (16.3% vs. 11.4%), but directional consistency is 100% (10/10 death-prone tasks in both models). `identify-life-stages-1` shows cross-model precision matching (55%/10%/+45 pp in both models). `find-non-living-thing` shows zero deaths in both models, consistent with its majority-class answer distribution.

Task	Int%	Det%	Δ pp	D / I / N
identify-life-stages-1	55	10	+45	9 / 0 / 11
freeze	45	0	+45	9 / 0 / 11
grow-fruit	45	5	+40	8 / 0 / 12
identify-life-stages-2	45	5	+40	8 / 0 / 12
measure-melting-point	50	15	+35	9 / 0 / 11
lifespan-longest	35	5	+30	7 / 1 / 12
lifespan-l-then-s	50	20	+30	6 / 0 / 14
chemistry-mix	45	20	+25	5 / 0 / 15
inclined-plane	45	20	+25	6 / 0 / 14
find-animal	40	25	+15	6 / 2 / 12
find-non-living-thing	0	0	0	10 / 0 / 10
Aggregate	**41.4**	**11.4**	**+30.0**	**83 / 3 / 134**

Table 16: Per-task Llama-3.1-8B cross-model replication (N=20 per task). D/I/N: deterministic-preferred / internal-preferred / no-preference. Sorted by Δ DR descending.

Cross-model and cross-environment summary. Table 17 consolidates replication results across model families and environments.

Model	Environment	n	Int DR	Det DR	ΔDR	DAF	DAF 95% CI	Signal
Qwen2.5-7B	ScienceWorld	245	42.9%	16.3%	+26.5	90.4%	[81.1, 99.0]	death
Llama-3.1-8B	ScienceWorld	220	41.4%	11.4%	+30.0	88.2%	[80.2, 95.1]	death
Gemma-2-9B-it	ScienceWorld	220	36.4%	14.1%	+22.3	89.7%	[79.5, 98.4]	death
Qwen2.5-72B	ScienceWorld	255	30.6%	2.75%	+27.8	92.4%	—	death
Qwen2.5-7B	ALFWorld	135	0%	0%	0	0%	—	info-asym
Llama-3.1-8B	ALFWorld	135	0%	0%	0	0%	—	info-asym
Qwen2.5-7B	MATH	795	N/A	N/A	N/A	N/A	N/A	none

Table 17: Cross-model and cross-environment replication summary. **ScienceWorld**: death dominance replicates across three model families in the 7–9B range and persists under 10× scaling (72B); DAF 95% CIs from non-clustered episode-level bootstrap (B=10,000). **ALFWorld**: DAF = 0% (no death mechanism); routing signal arises from information asymmetry. **MATH**: no routing signal without catastrophic failure. Signal column: diagnosed routing mechanism. DR = death rate. Gemma-2's wider CI reflects one zero-death task type (`find-non-living-thing`); excluding it yields DAF = 99.9% [99.7, 100.0].

N Failure Mode Analysis

All 105 internal-strategy deaths (100%) are triggered by the same action pattern: `focus on [object]`. In ScienceWorld, `focus on` is the answer-submission action (equivalent to "this is my final answer"); the LLM agent systematically misuses it as a general interaction or exploration command, and applies it to irrelevant targets (rooms, doors, inventory items). Each such execution submits an incorrect answer and triggers the −100 death penalty. Within death episodes, both strategies exhibit comparable `focus on` rates (∼5%);

the internal strategy's higher death count (105 vs. 40 for deterministic) reflects how often each strategy enters catastrophic states, not different action selection within those states.

Manual inspection identifies four failure categories: (A) no task awareness, where the agent executes `focus on` without reading the task; (B) task-aware but lacking action grounding, where the agent correctly identifies the objective but cannot map it to valid action primitives; (D) genuine attempts with execution failure; and (E) initial progress wasted by early errors. Category B dominates: agents demonstrate task comprehension but systematically fail to translate it into correct ScienceWorld actions. This taxonomy was produced by a single annotator; no inter-annotator agreement was computed. The categories are intended as qualitative illustrations, not as formal labels with reliability guarantees.

Representative examples. *Category A* (no task awareness): In a `chemistry-mix` episode, the agent immediately executes `focus on kitchen`, `focus on door`, `focus on inventory` without reading the task, dying on the first `focus on` call. *Category B* (comprehension without grounding): In a `lifespan-longest-lived` episode, the agent correctly identifies that it must compare animal lifespans and reasons about which animal lives longest, but then executes `focus on oak tree` (a non-animal object in the room) instead of navigating to find the target animal, triggering death.

Action distribution analysis. Figure 5 shows the action distribution in death versus non-death episodes for the internal strategy; both are dominated by `connect` actions, with `focus on` moderately more frequent in death episodes (5.4% vs. 3.4%). Crucially, Figure 7 reveals that within death episodes, internal and deterministic strategies exhibit comparable `focus on` rates (∼5%). The survival gap (105 vs. 40 deaths) therefore reflects how often each strategy enters catastrophic states, not how it behaves within those states. Among fatal `focus on` targets, `workshop` remains the most frequent (20.0%), followed by `inventory` (17.1%) and `kitchen` (10.5%), indicating a systematic object bias rather than random exploration. Figure 6 visualizes the temporal pattern: agents execute long sequences of `connect` actions before a terminal `focus on` triggers death, typically in the final 1–3 steps.

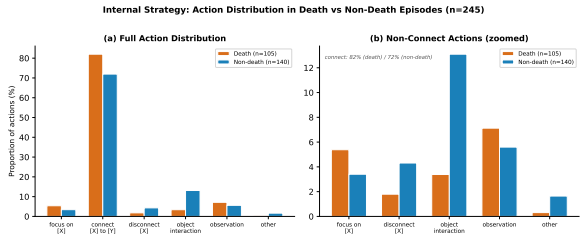


Figure 5: Action distribution for the internal (LLM) strategy in death (n=105) vs. non-death (n=140) episodes. Left: full distribution (both dominated by `connect`). Right: non-`connect` actions zoomed. `focus on` is moderately more frequent in death episodes (5.4% vs. 3.4%), while non-death episodes show higher rates of observation and object interaction.

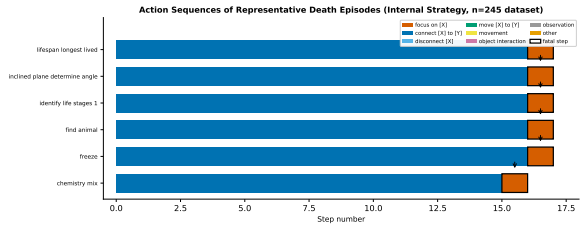


Figure 6: Step-by-step action sequences for 6 representative death episodes. The fatal `focus on` action (marked with arrow) consistently appears in the final steps after long `connect` sequences, and illustrates the "connect loop → fatal focus" pattern. Examples are drawn from the full 30-task dataset for task-type diversity; the pattern is identical in the 11-task analysis subset.

O Grounding Ablation

We test whether directly intervening on the dominant death action (`focus on`) eliminates death vulnerability on the 10 death-prone tasks (baseline 48.0%, n=200).

Prompt warning. Adding an explicit system prompt ("use `focus on` only for final answers") reduces aggregate death rate to 37.0% ($\Delta = -11$ pp). Both identification tasks exhibit ironic effects (`identify-life-stages-1`: 35%→50%; `identify-life-stages-2`: 25%→70%), where explicit mention of the dangerous action may have increased its salience.

Action-space filtering. Blocking `focus on` during the exploration phase yields 379 blocks across 200 episodes with zero leaks and zero passthroughs (the agent never persisted in retrying blocked actions), reducing death rate to 29.0%

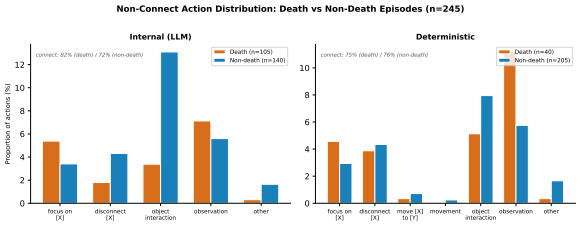


Figure 7: Non-`connect` action distribution comparison between internal (left) and deterministic (right) strategies. Within death episodes, both strategies exhibit comparable `focus on` rates (∼5%); the internal strategy's higher death count (105 vs. 40) reflects more frequent entry into catastrophic states, not different action selection within those states.

($\Delta = -19$ pp). The zero-passthrough rate suggests `focus on` use is a shallow behavioral pattern rather than a deliberate strategy.

Per-task variation. Under filtering, `identify-life-stages` and `inclined-plane` approach deterministic-level safety (∼15–20% DR), while `chemistry-mix` and `find-animal` remain at 35–40%. Residual deaths occur during the answer-submission window, where the filter is inactive and agents submit `focus on` to incorrect objects.

P Ablation Studies

Capacity scaling. We scale the implicit router from Qwen3-0.6B to Qwen3-4B and evaluate at both step and episode level (Table 18).

Model	Step Bal. Acc. (%)	Ep. Bal. Acc. (%)
Qwen3-0.6B	56.4 ± 3.5	50.7 ± 2.6
Qwen3-1.7B	58.2 ± 4.0	43.1 ± 1.0
Qwen3-4B	59.2 ± 3.5	$\mathbf{59.0 \pm 1.9}$
Always-det	50.0	50.0

Table 18: Capacity scaling: step-level and episode-level balanced accuracy (4 seeds, identical configuration across all scales). Step-level accuracy is flat (56–59%, within seed variance). At episode level, only 4B breaks the random baseline (59.0%, $p < 0.005$).

Step-level balanced accuracy is flat across scales (56–59%, within seed variance); step-level discrimination is not capacity-sensitive. At episode level, only 4B breaks the random baseline ($59.0 \pm 1.9\%$, $p < 0.005$); 0.6B and 1.7B remain at or below chance. The 1.7B model's below-chance episode-level accuracy (43.1%) may reflect overfitting to spurious step-level correlations that anti-correlate with episode outcomes; without per-class confusion matrices at this scale, this remains a hypothesis. The dominant explanation across scales is that death avoidance constitutes a task-type-level binary signal that a step-level router cannot exploit efficiently.

Generative router. A generative routing model trained with explicit routing labels achieves balanced accuracy of 46.7%, confirming that the routing bottleneck is signal-limited, not architecture-limited.

Leave-one-task-type-out routing. Under leave-one-task-type-out evaluation, blind routing collapses to always-deterministic (balanced accuracy 50.0%); a keyword-based variant achieves 62.9% only through task-identity leakage, not genuine routing signal. A guided router evaluated under the same protocol confirms this: without task-type identifiers it achieves chance accuracy (50.0%), and leaking task names raises this to 62.9%, confirming that the only learnable routing feature is task-type identity.

Death-proximal labeling. Training a router only on the k steps preceding death events (rather than all steps) also collapses to majority-class prediction (step balanced accuracy 50.0%, episode balanced accuracy 50.0%, internal recall 0%), confirming that the absence of step-level signal is not an artifact of label dilution across long episodes.

Reasoning depth. An ablation comparing chain-of-thought reasoning against direct-answer generation on 6 task types (n=120 episodes per condition, 6 tasks $\times$ 20 episodes per task) found no significant difference (Mann–Whitney $p > 0.26$, $|d| \leq 0.31$) and near-identical death rates (41.8% vs. 42.5%); reasoning depth does not meaningfully alter internal-mode outcomes.

Q Threshold Routing Analysis

Given per-task death risk knowledge, we test whether a simple threshold routing strategy can outperform always-deterministic. For each threshold $t \in [0, 1]$, we route to internal if the per-task internal death rate $\leq t$, otherwise to deterministic, and sweep all thresholds to find the optimal.

Even with perfect per-task death risk knowledge, threshold routing cannot outperform always-deterministic. The sole zero-death task achieves

Strategy	Mean Score	Death Rate	Δ vs Det
Always-det	−9.47	0.163	0.00
Optimal threshold (t=0)	−11.03	0.155	−1.56
Always-internal	−39.07	0.429	−29.60

Table 19: Death-risk threshold routing analysis on 245 episodes. The optimal threshold routes only `find-non-living-thing` (the sole zero-death task) to internal, with all other tasks routed to deterministic. Bootstrap 95% CI for the difference: $[-8.87, +5.92]$.

higher scores under deterministic (+28.5) than internal (+18.9), leaving no room for a beneficial mixed strategy.

R Practitioner Decision Framework

Given a new interactive environment, the following three-stage procedure diagnoses whether learned routing adds value over a fixed deterministic strategy.

Stage 1: Pilot collection (∼50 episodes). Run pure-internal and pure-deterministic strategies on a representative sample (∼50 episode pairs). Compute the death-rate differential $\Delta\text{DR} = \text{DR}_{\text{int}} - \text{DR}_{\text{det}}$.

Stage 2: Death dominance check. If $\Delta\text{DR} > 15$ pp, death asymmetry is likely the dominant routing signal. Compute the death-label fraction f_{death}: the proportion of non-tie routing labels where exactly one strategy dies.

Stage 3: Decision. If $f_{\text{death}} > 75\%$, the routing signal is death-dominated; a fixed deterministic strategy will match or exceed any learned router. If $f_{\text{death}} < 50\%$, diverse routing signals may justify a learned router. Between 50%–75%, collect additional episodes to reduce estimation noise before committing to router development.

Retrospective validation. All four experimental conditions in this paper are correctly classified:

Condition	ΔDR	f_{death}	Prediction	Actual
Qwen-7B	+26.5	78.9%	det-fixed	det wins
Llama-8B	+30.0	>75%	det-fixed	det wins
Qwen-72B	+27.8	>75%	det-fixed	det wins
MATH	+0.5	0%	router viable	no signal

Table 20: Retrospective validation of the decision framework. All four conditions are correctly classified (4/4).

The framework also predicts the inverted-U relationship between capability and death risk: below ∼0.5B parameters, agents rarely reach dangerous states (13.3% death rate in a 15-episode pilot), so ΔDR is low and the framework would not recommend fixed-deterministic. Above ∼7B, agents are capable enough to enter catastrophic states but not capable enough to avoid them, producing high ΔDR.

Threshold calibration. The thresholds above (15 pp, 75%, 50%) are illustrative, derived from ScienceWorld's specific reward structure (binary death penalty P=100, sparse terminal rewards). Environments with graded penalties, dense rewards, or qualitatively different failure modes may require different thresholds. Practitioners should calibrate these boundaries for their specific environment.

S Extended Validation of Excluded Tasks

To verify that task exclusion does not bias toward death-prone tasks, we collect n=20 episodes for five representative excluded tasks spanning all exclusion categories (Table 21).

Task	Category	n	Int DR	Det DR	ΔDR	D/I/N
boil	dev-set	20	55%	25%	+30 pp	6/0/14
melt	dev-set	20	45%	20%	+25 pp	7/2/11
find-living-thing	taxonomic	20	55%	15%	+40 pp	10/0/10
use-thermometer	agent-floor	20	50%	20%	+30 pp	7/1/12
power-component	agent-floor	20	50%	15%	+35 pp	7/0/13
Mean (5 tasks)		100	51.0%	19.0%	+32.0 pp	37/3/60

Table 21: Extended validation (n=20) of representative excluded task types. All five tasks confirm death asymmetry; mean $\Delta\text{DR} = +32.0$ pp is consistent with the n=5 prescreening estimate for these same tasks (Table 4).

Notably, the excluded-task aggregate DAF (0.991) exceeds the included-task DAF (0.904; Table 3), indicating that excluded tasks are *more* death-dominated than included tasks—the opposite of a cherry-picking bias.

T Agent Prompt Templates

Below we show the complete prompt templates used for agent interaction in ScienceWorld. Both the internal (ReAct) and deterministic strategies share the same system prompt but differ in the user message template.

System prompt (shared). Used for all ScienceWorld agent interactions across both computation modes.

```
You are an agent interacting with a
    ScienceWorld text
environment. Your goal is to
    complete the given task by
choosing actions from the available
    action list.

Use the following format for each
    step:
Thought: <your reasoning about what
    to do next>
Action: <exact action string from
    the valid actions list>

Rules:
- You MUST choose an action from the
     valid actions list.
  Do not invent actions.
- Think step by step about what
    physical/scientific
  process is needed.
- Pay attention to the current
    observation for clues
  about object states.
```

Internal (ReAct) user template. At each step, the agent receives the current observation, valid action list, and recent action history, and must select a single action.

```
Current observation:
{observation}

Valid actions:
{valid_actions_str}

Previous actions taken: {history_str}


Choose your next action.
```

Deterministic candidate generation template. For the deterministic strategy, the agent generates ranked candidate actions instead of selecting one; these candidates are then evaluated via simulator forking (§2).

```
Current observation:
{observation}

Valid actions:
{valid_actions_str}

Previous actions taken: {history_str}


Generate the top 3 most promising
    actions from the valid
actions list, ranked by how likely
    they are to make
progress on the task. Output exactly
     3 actions, one per
line, in this format:
1. <action>
2. <action>
3. <action>
```

Reflexion self-reflection prompt. After a failed episode, the Reflexion agent (§3.3) generates a reflection that is injected into subsequent trial attempts as additional system context.

```
You just completed an episode in a
    ScienceWorld text
environment but failed.

Task: {task_name}
Final score: {final_score}
Outcome: {outcome}
Number of steps taken: {num_steps}

Here is a summary of your actions:
{action_summary}

Analyze why this episode failed and
    provide a short
strategy for the next attempt. Focus
     on:
1. What went wrong (wrong sequence,
    missed key step,
   dangerous action)?
2. What should be done differently
    next time?

Keep your reflection under 200 words.
```

Grounding warning (ablation). The prompt warning condition in the grounding ablation (Appendix O) appends the following to the shared system prompt:

```
CRITICAL SAFETY RULE: The 'focus on
    [object]' action is
ONLY for submitting your final
    answer. Do NOT use
'focus on' to examine, inspect, or
    interact with objects
during exploration. Using 'focus on'
     on the wrong object
will cause irreversible failure. Use
     'look at',
'examine', or 'pick up' instead for
    exploration.
```

The key difference between the two computation modes is the user message: the internal strategy asks the LLM to *select* one action (ReAct), while the deterministic strategy asks it to *rank* candidates for simulator evaluation. Both use identical system prompts and identical observation formatting.